Supplementary Materials for A Formal Investigation of Paradoxes Between Latent Aptitude and Observed Achievement via Full House Methodology

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This supplementary materials includes proofs of technical results and additional numerical examples supporting conclusions from Section 4 the main manuscript.

1 Proof to Lemma 4.4

Proof. Notice that the cumulative distribution function of a multivariate normal vector admits a sequential representation in terms of conditional univariate normal distribution functions. In particular, for $\mathbf{W} = (W_1, \ldots, W_n)^\top \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, there exist conditional CDFs $F_j(\cdot)$ such that

$$\boldsymbol{\Phi}(\mathbf{w}, \boldsymbol{\Sigma}) = \prod_{j=1}^{n} F_j(w_1, \ldots, w_j), \tag{1}$$

where

$$F_1(w_1) := \Pr(W_1 \le w_1), \qquad F_j(w_1, \ldots, w_j) := \Pr(W_j \le w_j \mid W_1 = w_1, \ldots, W_{j-1} = w_{j-1}), \quad j \ge 2,$$

and each F_j is the CDF of a univariate normal distribution.

where $\mathbf{W} = (W_1, W_2, \ldots, W_n)$, $\mathbf{w} = (w_1, w_2, \ldots, w_n)$.

Then in general, for any $k \in \{2, \ldots, n_1 + n_2 - 1\}$, the conditional distribution of W_k given $W_1, \ldots, W_{k-1}$ is

$$W_k \mid W_1 = w_1, \ldots, W_{k-1} = w_{k-1} \sim \mathcal{N}\left(\mu_k, \sigma_k^2\right)$$

with

$$\mu_k = \boldsymbol{\Sigma}_{k,1:(k-1)} \boldsymbol{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \mathbf{w}_{1:(k-1)}$$

$$\sigma_k^2 = \boldsymbol{\Sigma}_{k,k} - \boldsymbol{\Sigma}_{k,1:(k-1)} \boldsymbol{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \boldsymbol{\Sigma}_{1:(k-1),k}$$

Here: $\boldsymbol{\Sigma}_{k,1:(k-1)}$ is a $1 \times (k-1)$ row vector: the covariances between W_k and $W_1, \ldots, W_{k-1}$ $\boldsymbol{\Sigma}_{1:(k-1),1:(k-1)}$ is a $(k-1) \times (k-1)$ covariance matrix, $\boldsymbol{\Sigma}_{k,k}$ is the kth element on the diagonal of the covariance matrix $\boldsymbol{\Sigma}$, and $\mathbf{w}_{1:(k-1)} = (w_1, \ldots, w_{k-1})^T$.

Then we will show the the parameters μ and σ^2 of the conditional univariate normal distribution in the set of Equation (20) are finite.

When $k \leq n_1$, by applying the Equation (19)

$$
\begin{aligned}
\mu_k &= \boldsymbol{\Sigma}_{k,1:(k-1)} \boldsymbol{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \mathbf{w}_{1:(k-1)} \\
&= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}(k-2)+1 & -\frac{\sigma_2^2}{\sigma_1^2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2} \\ -\frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2} \\ -\frac{\sigma_2^2}{\sigma_1^2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}(k-2)+1 \end{bmatrix}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\frac{\sigma_2^2(\mu_2-\mu_1)}{\sigma_1^3}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}
$$

$$
\begin{aligned}
\sigma_k^2 &= \boldsymbol{\Sigma}_{k,k} - \boldsymbol{\Sigma}_{k,1:(k-1)} \boldsymbol{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \boldsymbol{\Sigma}_{1:(k-1),k} \\
&= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{\frac{\sigma_2^4}{\sigma_1^4}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}
$$

Then

$$
\begin{aligned}
&\Phi(W_k \le w_k | W_1 = w_1, W_2 = w_2, \ldots, W_{k-1} = w_{k-1}) \\
&= \Phi\left(\frac{\frac{\mu_2-\mu_1}{\sigma_1} - \frac{\frac{\sigma_2^2(\mu_2-\mu_1)}{\sigma_1^3}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}{\sqrt{\frac{\sigma_2^2}{\sigma_1^2}+1-\frac{\frac{\sigma_2^4}{\sigma_1^4}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right) \\
&= \Phi\left(\frac{\frac{\mu_2-\mu_1}{\sigma_1}(1-\frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1})}{\sqrt{1+\frac{\sigma_2^2}{\sigma_1^2}(1-\frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1})}} \right) \\
&= \Phi\left(\frac{\frac{\frac{\mu_2-\mu_1}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}{\sqrt{1+\frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right) \\
&= \Phi\left(\frac{\frac{\sigma_1(\mu_2-\mu_1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right) \\
&= c_k
\end{aligned}
$$

$$
\lim_{k\to+\infty} c_k = \lim_{k\to+\infty} \Phi\left(\frac{\frac{\sigma_1(\mu_2-\mu_1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right) = \Phi(0) = 0.5
$$

where $\Phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, for $k \le n_1$, $c_k < 1$.

When $k = n_1 + 1$, the only difference from the case $k \le n_1$ is that $\mathbf{\Sigma}_{k,\,1:(k-1)}$ is different. We can then apply a similar method as in the case $k \le n_1$ to compute μ_k and σ_k.

$$
\begin{aligned}
\mu_k &= \mathbf{\Sigma}_{k,1:(k-1)} \mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \mathbf{w}_{1:(k-1)} \\
&= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}^T_{(k-1)\times 1}
\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)}
\begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1}
\end{aligned}
$$

$$= \frac{\frac{\sigma_2(\mu_2-\mu_1)}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}$$

$$\begin{aligned}
\sigma_k^2 &= \mathbf{\Sigma}_{k,k} - \mathbf{\Sigma}_{k,1:(k-1)}\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}\mathbf{\Sigma}_{1:(k-1),k} \\
&= 2 - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= 2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}$$

Then

$$\begin{aligned}
&\Phi(W_k \le w_k | W_1 = w_1, W_2 = w_2, \dots, W_{k-1} = w_{k-1}) \\
&= \Phi\left(\frac{0 - \frac{\frac{\sigma_2(\mu_2-\mu_1)}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}{\sqrt{2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right) \\
&= \Phi\left(-\frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right) \\
&= \Phi\left(-\frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right) \\
&= c_k
\end{aligned}$$

$$\lim_{k\to+\infty} c_k = \lim_{k\to+\infty} \Phi\left(-\frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right) = \Phi\left(-\frac{\mu_2-\mu_1}{\sigma_2} \right)$$

where $\phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, for $k = n_1 + 1$, $c_k < 1$.

When $k > n_1 + 1$, we first compute $\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}$. Based on the structure of $\mathbf{\Sigma}$ in Equation (14), the $\mathbf{\Sigma}_{1:(k-1),1:(k-1)}$ can be decomposed into

$$\boldsymbol{\Sigma}_{1:(k-1),1:(k-1)} = \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix} \tag{2}$$

defined with the following structure

$$D_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{n_1 \times n_1} \quad D_{12}^T = D_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(k-n_1-1)\times n_1}$$

$$D_{22} = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(k-n_1-1)\times(k-n_1-1)} . \tag{3}$$

By applying the Equation (19) and Schur complement [Zhang, 2006],

$$\boldsymbol{\Sigma}^{-1}_{1:(k-1),1:(k-1)} = \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}, \tag{4}$$

$$E_{11} = (D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \cdot \frac{1}{k-n_1} \cdot \begin{bmatrix} k-n_1-1 & -1 & \dots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \dots & -1 & k-n_1-1 \end{bmatrix} \right.$$

$$\left. \cdot \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1} =$$

$$
\begin{aligned}
&= \begin{bmatrix} \frac{\sigma_2^2}{(k-n_1)\sigma_1^2}+1 & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \cdots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} \\ \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} \\ \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \cdots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2}+1 \end{bmatrix}^{-1} \\
&= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}_{n_1\times n_1}
\end{aligned}
$$

$$
\begin{aligned}
E_{12} &= -(D_{11}-D_{12}D_{22}^{-1}D_{21})^{-1}D_{12}D_{22}^{-1} \\
&= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix} \\
&\quad \times \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \cdots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & 2 \end{bmatrix}^{-1} \\
&= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix} \\
&\quad \times \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{k-n_1} \begin{bmatrix} k-n_1-1 & -1 & \cdots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -1 & k-n_1-1 \end{bmatrix} \\
&= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}
\end{aligned}
$$

$$\times \begin{bmatrix} \frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}_{n_1\times(k-n_1-1)}$$

Since $\mathbf{\Sigma}_{1:(k-1),1:(k-1)}$ is a symmetric matrix, then $E_{12} = E_{21}^T$. Thus,

$$E_{21} = -D_{22}^{-1}D_{21}(D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}_{(k-n_1-1)\times n_1}$$

$$E_{22} = D_{22}^{-1} + D_{22}^{-1}D_{21}(D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}D_{12}D_{22}^{-1}$$

$$= \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}^{-1} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}^{-1}$$

$$= \frac{1}{k-n_1} \begin{bmatrix} k-n_1-1 & -1 & \ldots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \ldots & -1 & k-n_1-1 \end{bmatrix}$$

$$
+\begin{bmatrix}
\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\
\vdots & \ddots & & \vdots \\
\vdots & & \ddots & \vdots \\
\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}
\end{bmatrix}
\begin{bmatrix}
\frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \\
\vdots & \ddots & & \vdots \\
\vdots & & \ddots & \vdots \\
\frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1}
\end{bmatrix}
$$

$$
= \frac{1}{k-n_1}\begin{bmatrix}
k-n_1-1 & -1 & \dots & -1 \\
-1 & \ddots & \ddots & \vdots \\
\vdots & \ddots & \ddots & -1 \\
-1 & \dots & -1 & k-n_1-1
\end{bmatrix}
+\begin{bmatrix}
\frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\
\vdots & \ddots & & \vdots \\
\vdots & & \ddots & \vdots \\
\frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}
\end{bmatrix}
$$

$$
= \begin{bmatrix}
1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\
-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \ddots & \ddots & \vdots \\
\vdots & \ddots & \ddots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\
-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}
\end{bmatrix}_{(k-n_1)\times(k-n_1)}
$$

Then

$$
\begin{aligned}
\mu_k &= \mathbf{\Sigma}_{k,1:(k-1)}\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}\mathbf{w}_{1:(k-1)} \\
&= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}^T_{(k-1)\times 1}
\begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}_{(k-1)\times(k-1)}
\begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{(k-1)\times 1} \\
&= \left[\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \quad , \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{12} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{22} \right]
\begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}
\end{aligned}
$$

$$= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix} - \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}n_1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} - \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1(k-n_1-1)}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}$$

$$= \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}$$

$$\sigma_k^2 = \mathbf{\Sigma}_{k,k} - \mathbf{\Sigma}_{k,1:(k-1)}\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}\mathbf{\Sigma}_{1:(k-1),k}$$

$$= 2 - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}_{(k-1)\times 1}$$

$$= 2 - \left[\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \quad , \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{12} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{22} \right] \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$
\begin{aligned}
&= 2 - \Bigg(\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} \\
&+ \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{12} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{22} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \Bigg) \\
&= 2 - \Bigg(\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} \\
&+ \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} \\
&+ \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \\
&+ \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \Bigg) \\
&= 2 - \Bigg(\begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} - \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} \\
&- \begin{bmatrix} \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} 1-\frac{\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & 1-\frac{\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \Bigg)
\end{aligned}
$$

$$= 2 - \left(\frac{\frac{\sigma_2^2}{\sigma_1^2} n_1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1} + 1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{2(k-n_1-1)n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1} + 1} + \frac{k-n_1-1}{k-n_1} + \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{(k-n_1-1)^2 n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1} + 1}\right)$$

$$= 2 - \frac{k-n_1}{k-n_1+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1} + 1}$$

Then

$$\begin{aligned}
&\Phi(W_k \le w_k | W_1 = w_1, W_2 = w_2, \ldots, W_{k-1} = w_{k-1}) \\
&= \Phi\left(\frac{0 - \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}}\right) \\
&= \Phi\left(-\frac{\frac{\frac{(\mu_2-\mu_1)\sigma_2}{\sigma_1^2}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}}\right) \\
&= \Phi\left(-\frac{\frac{(\mu_2-\mu_1)\sigma_2 n_1}{\sigma_2^2 n_1 + \sigma_1^2(k-n_1)}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\sigma_2^2 n_1}{\sigma_2^2 n_1(k-n_1) + \sigma_1^2(k-n_1)^2}}}\right) \\
&= c_k
\end{aligned}$$

$$\lim_{k\to+\infty} c_k = \lim_{k\to+\infty} \Phi\left(-\frac{\frac{(\mu_2-\mu_1)\sigma_2 n_1}{\sigma_2^2 n_1 + \sigma_1^2(k-n_1)}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\sigma_2^2 n_1}{\sigma_2^2 n_1(k-n_1) + \sigma_1^2(k-n_1)^2}}}\right) = \Phi(0) = 0.5$$

where $\phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, for all k, $c_k < 1$. Moreover, we show the parameters μ and σ^2 of the conditional univariate normal distribution appearing in Equation (20) are finite.

Now

$$\begin{aligned}
c = \max\Big\{ &\mathrm{P}(W_{n_1+n_2-1} \le w_{n_1+n_2-1} \,|\, W_1 = w_1, \ldots, W_{n_1+n_2-2} = w_{n_1+n_2-2}), \\
&\mathrm{P}(W_{n_1+n_2-2} \le w_{n_1+n_2-2} \,|\, W_1 = w_1, \ldots, W_{n_1+n_2-3} = w_{n_1+n_2-3}), \\
&\ldots, \mathrm{P}(W_2 \le w_2 \mid W_1 = w_1), \mathrm{P}(W_1 \le w_1)\Big\}, \\
= \max\Big\{ &c_2, c_3, \ldots, c_{n_1+n_2-1}, \mathrm{P}(W_1 \le w_1)\Big\}
\end{aligned} \tag{5}$$

Since all terms in the set above are the cumulative distribution functions of univariate normal distribution and $\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \ldots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \ldots, 0)$, then c is strictly less than 1.

$$\mathbf{\Phi}(\mathbf{W}, \mathbf{\Sigma}) \leq c^{n_1+n_2-1} \tag{6}$$

□

2 Proof to Theorem 4.5

Proof. Let $\hat{\mu}_1$, $\hat{\mu}_2$, $\hat{\sigma}_1$, and $\hat{\sigma}_2$ be consistent estimators of the population parameters μ_1, μ_2, σ_1, and σ_2, respectively, as defined in Section 4.2. By replacing the population parameters in $\mathbf{w}$ and $\mathbf{\Sigma}$ with these estimators, we obtain the sample-level analogs $\hat{\mathbf{w}}$ and $\hat{\mathbf{\Sigma}}$. Specifically, we define:

$$\hat{\mathbf{w}} = \left(\frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_1}, \ldots, \frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_1}, 0, \ldots, 0 \right) \in \mathbb{R}^{n_1+n_2-1},$$

where the first n_1 entries correspond to the standardized mean difference, and the remaining $n_2 - 1$ entries are zero, reflecting the structure introduced in Section 4.2.

The estimated covariance matrix $\hat{\mathbf{\Sigma}}$ has the block form:

$$\hat{\mathbf{\Sigma}} = \begin{bmatrix} \hat{\Sigma}_{11} & \hat{\Sigma}_{12} \\ \hat{\Sigma}_{21} & \hat{\Sigma}_{22} \end{bmatrix},$$

where each submatrix is obtained by replacing the population parameters in $\mathbf{\Sigma}$ with their respective sample estimates. These submatrices retain the same structural forms as defined in Section 4.2.

Based on the Equation (16), we can rewrite

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \hat{\mu_1}.\hat{\mu_2}, \hat{\sigma_1}, \hat{\sigma_2}\right) - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \mu_1.\mu_2, \sigma_1, \sigma_2\right) \\ &= n_2(\Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma)) \end{aligned}$$

Consider the term $\Phi(\hat{\mathbf{w}}, \hat{\mathbf{\Sigma}}) - \Phi(\mathbf{w}, \mathbf{\Sigma})$. We perform a Taylor expansion of this expression with respect to the parameters μ_1, μ_2, σ_1, and σ_2.

$$\begin{aligned} \Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma) \approx &\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}(\hat{\mu_1} - \mu_1) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_2}(\hat{\mu_2} - \mu_2) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_1}(\hat{\sigma_1} - \sigma_1) \\ &+ \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_2}(\hat{\sigma_2} - \sigma_2) \end{aligned}$$

Then for the first derivative term $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}$:

$$\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1} = \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \cdot \frac{\partial w_i}{\partial \mu_1}$$

$$= -\frac{1}{\sigma_1}\sum_{i=1}^{n_1}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_i}$$

By applying Leibniz rule and without loss of generality, we compute the $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_1}$

$$\begin{aligned}
\frac{\partial\Phi(\mathbf{w};\mathbf{\Sigma})}{\partial w_1} &= \frac{\partial}{\partial w_1}\int_{-\infty}^{w_1}\cdots\int_{-\infty}^{w_{n_1+n_2-1}} f_{\mathbf{W}}(z_1,\ldots,z_{n_1+n_2-1})dz_{n_1+n_2-1}\ldots dz_1 \\
&= \int_{-\infty}^{w_2}\cdots\int_{-\infty}^{w_{n_1+n_2-1}} f_{\mathbf{W}}(w_1,z_2,\ldots,z_{n_1+n_2-1})dz_{n_1+n_2-1}\ldots dz_2 \\
&= \int_{-\infty}^{w_2}\cdots\int_{-\infty}^{w_{n_1+n_2-1}} f_{W_1}(w_1)\cdot f_{\tilde{\mathbf{W}}}(\tilde{w})dz_{n_1+n_2-1}\ldots dz_2 \\
&= \phi(w_1)\Phi(\tilde{\mathbf{w}},\tilde{\Sigma})
\end{aligned}$$

where $\phi(w_1)$ is the probability density function of univariate normal distribution with $\mu = 0$ and $\sigma^2 = \frac{\sigma_2^2}{\sigma_1^2} + 1$, $\Phi(\tilde{\mathbf{w}},\tilde{\Sigma})$ is the conditional cumulative distribution function of the remaining components given $W_1 = w_1$. By applying Schur complement [Zhang, 2006],

$$\tilde{\mathbf{w}} = (w_2,\ldots,w_{n_1+n_2-1}) - (\frac{\sigma_2^2}{\sigma_1^2},\ldots,\frac{\sigma_2^2}{\sigma_1^2},\frac{\sigma_2}{\sigma_1},\ldots,\frac{\sigma_2}{\sigma_1}) * (\frac{\sigma_2^2}{\sigma_1^2}+1)^{-1} * (w_1)$$

$\tilde{\mathbf{w}}$ is a vector with the first $n_1 - 1$ elements is $\frac{\mu_2-\mu_1}{\sigma_1} - \frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}$ and the last $n_2 - 1$ elements are $-\frac{(\mu_2-\mu_1)\sigma_2}{\sigma_1^2+\sigma_2^2}$.

$$\tilde{\mathbf{\Sigma}} = \mathbf{\Sigma}_{-1,-1} - \mathbf{\Sigma}_{-1,1}\mathbf{\Sigma}_{1,1}^{-1}\mathbf{\Sigma}_{1,-1}$$

where $\mathbf{\Sigma}_{-1,-1}$ denotes the submatrix of $\mathbf{\Sigma}$ obtained by removing the first row and the first column, $\mathbf{\Sigma}_{-1,1}$ denotes the column vector obtained by removing the first element from the first column of $\mathbf{\Sigma}$, $\mathbf{\Sigma}_{1,-1}$ denotes the row vector obtained by removing the first element from the first row of $\mathbf{\Sigma}$, and $\mathbf{\Sigma}_{1,1}$ denotes the top-left scalar element of $\mathbf{\Sigma}$.

Let denote the form of the matrix $\tilde{\mathbf{\Sigma}}$:

$$\tilde{\mathbf{\Sigma}} = \begin{bmatrix}\tilde{\Sigma}_{11} & \tilde{\Sigma}_{12}\\ \tilde{\Sigma}_{21} & \tilde{\Sigma}_{22}\end{bmatrix}$$

$$\tilde{\Sigma}_{11} = \begin{bmatrix}\frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2}\\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2}\\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1\end{bmatrix}_{(n_1-1)\times(n_1-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}+1}\begin{bmatrix}\frac{\sigma_2^4}{\sigma_1^4} & \cdots & \cdots & \frac{\sigma_2^4}{\sigma_1^4}\\ \vdots & \ddots & & \vdots\\ \vdots & & \ddots & \vdots\\ \frac{\sigma_2^4}{\sigma_1^4} & \cdots & \cdots & \frac{\sigma_2^4}{\sigma_1^4}\end{bmatrix}_{(n_1-1)\times(n_1-1)}$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_1-1)\times(n_1-1)}$$

$$\tilde{\Sigma}_{12}^T = \tilde{\Sigma}_{21} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(n_2-1)\times(n_1-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}+1} \begin{bmatrix} \frac{\sigma_2^3}{\sigma_1^3} & \cdots & \cdots & \frac{\sigma_2^3}{\sigma_1^3} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^3}{\sigma_1^3} & \cdots & \cdots & \frac{\sigma_2^3}{\sigma_1^3} \end{bmatrix}_{(n_2-1)\times(n_1-1)}$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_2-1)\times(n_1-1)}$$

$$\tilde{\Sigma}_{22} = \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}_{(n_2-1)\times(n_2-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

$$= \begin{bmatrix} 2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & 2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

Therefore, $\Phi(\tilde{\mathbf{w}}, \tilde{\Sigma})$ is finite. Then for $i = 1, \ldots, n_1$, $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial w_i}$ is the same as $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial w_1}$. By applying Lemma 4.4, we get

$$\begin{aligned}
\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1} &= -\frac{1}{\sigma_1} \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \\
&= -\frac{n_1}{\sigma_1} \cdot \phi(w_1) \cdot \Phi(\tilde{\mathbf{w}}, \tilde{\Sigma}) \\
&\leq O_P(n_1 c^{n_1+n_2-1})
\end{aligned}$$

By applying the similar method used for $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \mu_1}$, we have

$$\begin{aligned}
\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_2} &= \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \cdot \frac{\partial w_i}{\partial \mu_2} \\
&= \frac{1}{\sigma_1} \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \\
&= \frac{n_1}{\sigma_1} \cdot \phi(w_1) \cdot \Phi(\tilde{\mathbf{w}}, \tilde{\Sigma}) \\
&= OP(n_1 c^{n_1+n_2-1})
\end{aligned}$$

Notice that $\Phi(\mathbf{w}, \Sigma)$ has two parameters $\mathbf{w}$, and Σ involving σ_1, then using the Equation (137) in Petersen et al. [2008] and the similar method used for $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \mu_1}$, we have

$$\begin{aligned}
\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_1} &= \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \cdot \frac{\partial w_i}{\partial \sigma_1} + \operatorname{tr}(\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1}) \\
&= -\frac{(\mu_2 - \mu_1) n_1}{\sigma_1^2} \phi(w_1) \cdot \Phi(\tilde{\mathbf{w}}, \tilde{\Sigma}) + \operatorname{tr}(\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1})
\end{aligned}$$

Before computing $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma}$, we define the $\phi(\mathbf{x})$ be the probability density function of $\Phi(\mathbf{w}, \Sigma)$, which is

$$\phi(\mathbf{x}) = \frac{1}{\sqrt{\det(2\pi\Sigma)}} \exp\left[-\frac{1}{2}(\mathbf{x})^T \Sigma^{-1}(\mathbf{x})\right]$$

then

$$\log(\phi(\mathbf{x})) = -\frac{1}{2}(\log(\det(2\pi\Sigma)) + \mathbf{x}^t \Sigma^{-1} \mathbf{x})$$

then using the Equation (57) and (61) in Petersen et al. [2008], we have

$$\frac{\partial \log(\phi(\mathbf{x}))}{\partial \Sigma} = -\frac{1}{2}(\frac{\partial \log(\det(2\pi\Sigma))}{\partial \Sigma} + \frac{\partial \mathbf{x}^t \Sigma^{-1} \mathbf{x}}{\partial \Sigma})$$

$$= -\frac{1}{2}(\Sigma^{-1} - \Sigma^{-1}\mathbf{x}\mathbf{x}^t\Sigma^{-1})$$

Then for the term $\text{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma} \cdot \frac{\partial\Sigma}{\partial\sigma_1})$ is:

$$\begin{aligned}
\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma} &= \frac{\partial\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})d\mathbf{x}}{\partial\Sigma} \\
&= \int_{-\infty}^{\mathbf{w}}\frac{\partial\phi(\mathbf{x})}{\partial\Sigma}d\mathbf{x} \\
&= \int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\frac{\partial\log(\phi(\mathbf{x}))}{\partial\Sigma}d\mathbf{x} \\
&= -\frac{1}{2}\Sigma^{-1}\Phi(\mathbf{w},\Sigma) + \frac{1}{2}\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} \\
&= \frac{1}{2}\Sigma^{-1}\left(\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} - \Phi(\mathbf{w},\Sigma)\right)
\end{aligned}$$

Then

$$\begin{aligned}
\text{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma} \cdot \frac{\partial\Sigma}{\partial\sigma_1}) &= \text{tr}\left(\frac{1}{2}\Sigma^{-1}\left(\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} - \Phi(\mathbf{w},\Sigma)\right)\frac{\partial\Sigma}{\partial\sigma_1}\right) \\
&= \frac{1}{2}\,\text{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) - \frac{1}{2}\,\text{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_1}\right)
\end{aligned}$$

To compute the $\text{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma} \cdot \frac{\partial\Sigma}{\partial\sigma_1})$, we firstly compute the $\text{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_1}\right)$. Let define $\frac{\partial\Sigma}{\partial\sigma_1}$ be:

$$\begin{aligned}
\frac{\partial\Sigma}{\partial\sigma_1} &= \begin{bmatrix}\frac{\partial\Sigma_{11}}{\partial\sigma_1} & \frac{\partial\Sigma_{12}}{\partial\sigma_1} \\ \frac{\partial\Sigma_{21}}{\partial\sigma_1} & \frac{\partial\Sigma_{22}}{\partial\sigma_1}\end{bmatrix} \\
&= \begin{bmatrix}A_{11} & A_{12} \\ A_{21} & A_{22}\end{bmatrix}
\end{aligned}$$

$$A_{11} = \left[-\frac{2\sigma_2^2}{\sigma_1^3}\right]_{n_1\times n_1}, A_{12}^T = A_{21} = \left[-\frac{\sigma_2}{\sigma_1^2}\right]_{(n_2-1)\times n_1}, A_{22} = \left[0\right]_{(n_2-1)\times(n_2-1)}$$

and by applying Schur complement [Zhang, 2006], we have

$$\mathbf{\Sigma}^{-1} = \begin{bmatrix}\Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22}\end{bmatrix}^{-1} = \begin{bmatrix}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} & -(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1} \\ -\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} & \Sigma_{22}^{-1} + \Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}\end{bmatrix}$$

where Σ_{11}, Σ_{12}, Σ_{21}, and Σ_{22} are defined in Equation (14).
By applying Equation (19), we have

$$\mathbf{\Sigma}^{-1} = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$B_{11} = (\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \dots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \dots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - (1-\frac{1}{n_2}) \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \dots & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \dots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix} \right)^{-1}$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{n_2\sigma_1^2}+1 & \frac{\sigma_2^2}{n_2\sigma_1^2} & \dots & \frac{\sigma_2^2}{n_2\sigma_1^2} \\ \frac{\sigma_2^2}{n_2\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{n_2\sigma_1^2} \\ \frac{\sigma_2^2}{n_2\sigma_1^2} & \dots & \frac{\sigma_2^2}{n_2\sigma_1^2} & \frac{\sigma_2^2}{n_2\sigma_1^2}+1 \end{bmatrix}^{-1} = \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix}_{n_1\times n_1}$$

$B_{12} = -(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \dots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \dots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix}\begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix}_{n_1\times(n_2-1)}$$

$B_{21} = -\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}$

$$= -\begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1}\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix}$$

$$= -\begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix}\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix}_{(n_2-1)\times n_1}$$

$B_{22} = \Sigma_{22}^{-1} + \Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}$

$$= \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix}\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}\begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1}$$

$$
= \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix}
$$

$$
= \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} + \begin{bmatrix} \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \end{bmatrix}
$$

$$
= \begin{bmatrix} 1-\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & 1-\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \end{bmatrix}_{(n_2-1)\times(n_2-1)}
$$

Then we have

$$
\Phi(\mathbf{w}, \Sigma) \leq c^{n_1+n_2-1}
$$

Therefore,

$$
\Sigma^{-1}\Phi(\mathbf{w}, \Sigma)\frac{\partial \Sigma}{\partial \sigma_1} \leq \Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_1}c^{n_1+n_2-1}
$$

Based on the structure of the Σ^{-1} and $\frac{\partial \Sigma}{\partial \sigma_1}$, we have

$$
\begin{aligned}
& \operatorname{tr}(\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_1}) \\
&= \operatorname{tr}\left(\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}\right) \\
&= \operatorname{tr}(B_{11}A_{11} + B_{12}A_{21}) + tr(B_{21}A_{12} + B_{22}A_{22}) \\
&= n_1\left(\frac{-\frac{2\sigma_2^2}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1-(n_1-1)\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2}) + \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\frac{\sigma_2}{\sigma_1^2}(n_2-1)\right)
\end{aligned}
$$

$$+ (n_2 - 1) \left(n_1 \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} \frac{\sigma_2}{\sigma_1^2} \right)$$
$$= -2 \frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1}$$

Now we have

$$\mathrm{tr}(\Sigma^{-1}\Phi(\mathbf{w}, \Sigma)\frac{\partial \Sigma}{\partial \sigma_1}) \leq 2\frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} c^{n_1+n_2-1}$$

Now we compute the term $\mathrm{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} \Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_1}\right)$. Let rewrite the $\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}$ into matrix form:

$$\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} = \begin{bmatrix} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1^2 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1 x_n d\mathbf{x} \\ \vdots & \ddots & \vdots \\ \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n x_1 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n^2 d\mathbf{x} \end{bmatrix}$$

Define $\mathbf{w}_{-i}$ as the vector $\mathbf{w}$ with the ith component removed, and $\mathbf{w}_{-(i,j)}$ as the vector $\mathbf{w}$ with the ith and j th components removed. Similarly, $\mathbf{x}_{-i}$ and $\mathbf{x}_{-(i,j)}$ are defined in the same manner.

Then

$$\begin{aligned} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_i^2 d\mathbf{x} &= \int_{-\infty}^{w_i} \left(\int_{-\infty}^{\mathbf{w}_{-i}} \phi(\mathbf{x}_{-i}|x_i) d\mathbf{x}_{-i} \right) \; x_i^2 \; \phi(x_i) dx_i \\ &\leq c^{n_1+n_2-3} \cdot Var(X_i) \end{aligned}$$

$$\begin{aligned} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_i x_j d\mathbf{x} &= \int_{-\infty}^{w_i} \int_{-\infty}^{w_j} \left(\int_{-\infty}^{\mathbf{w}_{-(i,j)}} \phi(\mathbf{x}_{-(i,j)}|x_i, x_j) d\mathbf{x}_{-(i,j)} \right) \; x_i x_j \; \phi(x_i, x_j) dx_i dx_j \\ &\leq c^{n_1+n_2-3} \cdot Cov(X_i, X_j) \end{aligned}$$

Replace $Var(X_i)$ and $Cov(X_i, X_j)$ using the elements of Σ, we have

$$\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} = \begin{bmatrix} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1^2 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1 x_n d\mathbf{x} \\ \vdots & \ddots & \vdots \\ \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n x_1 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n^2 d\mathbf{x} \end{bmatrix}$$

$$\leq \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

where

$$S_{11} = \begin{bmatrix} c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}+1) & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \dots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) \\ c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) \\ c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \dots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}+1) \end{bmatrix}_{n_1 \times n_1}$$

$$S_{12}^T = S_{21} = \begin{bmatrix} c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) & \dots & \dots & c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) & \dots & \dots & c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) \end{bmatrix}_{(n_2-1)\times n_1}$$

$$S_{22} = \begin{bmatrix} 2c^{n_1+n_2-3} & c^{n_1+n_2-3} & \dots & c^{n_1+n_2-3} \\ c^{n_1+n_2-3} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & c^{n_1+n_2-3} \\ c^{n_1+n_2-3} & \dots & c^{n_1+n_2-3} & 2c^{n_1+n_2-3} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

Then,

$$\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} = c^{n_1+n_2-3} * \Sigma$$

Therefore, followed by the results before,

$$\begin{aligned} &\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} \Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_1}\right) \\ &\leq \operatorname{tr}\left(\Sigma^{-1} \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_1}\right) \\ &= \operatorname{tr}\left(\Sigma^{-1} c^{n_1+n_2-3} * \Sigma * \Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_1}\right) \\ &= c^{n_1+n_2-3} \operatorname{tr}\left(\Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_1}\right) \\ &= 2\frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} c^{n_1+n_2-3} \end{aligned}$$

Then

$$\operatorname{tr}(\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1}) = \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} \Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_1}\right) - \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial \Sigma}{\partial \sigma_1}\right)$$

$$= O_p(c^{n_1+n_2-3})$$

By applying the similar method used for $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \mu_1}$, we have

$$\begin{aligned}\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_1} &= -\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma}) + \operatorname{tr}(\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma}\cdot\frac{\partial \Sigma}{\partial \sigma_1}) \\ &= -\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma}) + O_p(c^{n_1+n_2-3}) \\ &= O_P(n_1 c^{n_1+n_2-3})\end{aligned}$$

Similarly, we compute $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_2}$,

$$\begin{aligned}\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_2} &= \operatorname{tr}(\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma}\cdot\frac{\partial \Sigma}{\partial \sigma_2}) \\ &= \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}\right) - \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial \Sigma}{\partial \sigma_2}\right)\end{aligned}$$

To compute $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_2}$, firstly we have

$$\begin{aligned}\frac{\partial \Sigma}{\partial \sigma_2} &= \begin{bmatrix}\frac{\partial \Sigma_{11}}{\partial \sigma_1} & \frac{\partial \Sigma_{12}}{\partial \sigma_1} \\ \frac{\partial \Sigma_{21}}{\partial \sigma_1} & \frac{\partial \Sigma_{22}}{\partial \sigma_1}\end{bmatrix} \\ &= \begin{bmatrix}C_{11} & C_{12} \\ C_{21} & C_{22}\end{bmatrix}\end{aligned}$$

$$C_{11} = \left[\frac{2\sigma_2}{\sigma_1^2}\right]_{n_1\times n_1}, C_{12}^T = C_{21} = \left[\frac{1}{\sigma_1}\right]_{(n_2-1)\times n_1}, C_{22} = \left[0\right]_{(n_2-1)\times(n_2-1)}$$

Then by applying Lemma 4.4, we have

$$\Phi(\mathbf{w},\Sigma) \le c^{n_1+n_2-1}$$

Then

$$\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial \Sigma}{\partial \sigma_2} \le \Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}c^{n_1+n_2-1}$$

Based on the structure of the Σ^{-1} and $\frac{\partial \Sigma}{\partial \sigma_2}$, we have

$$\operatorname{tr}(\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2})$$

$$
\begin{aligned}
&= \operatorname{tr}\left(\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}\right) \\
&= \operatorname{tr}(B_{11}C_{11} + B_{12}C_{21}) + tr(B_{21}C_{12} + B_{22}C_{22}) \\
&= n_1 \left(\frac{\frac{2\sigma_2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1}\left(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1 - 1}{n_2} + 1 - (n_1 - 1)\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2}\right) \right) - \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1}\frac{1}{\sigma_1} n_1(n_2 - 1) \\
&- (n_2 - 1)\left(n_1 \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1}\frac{1}{\sigma_1} \right) \\
&= \frac{\frac{2\sigma_2}{\sigma_1^2}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1}
\end{aligned}
$$

Now we have

$$
\operatorname{tr}(\Sigma^{-1}\Phi(\mathbf{w}, \Sigma)\frac{\partial \Sigma}{\partial \sigma_2}) \leq \frac{\frac{2\sigma_2}{\sigma_1^2}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} c^{n_1+n_2-1}
$$

$$
\begin{aligned}
&\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} \Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}\right) \leq \operatorname{tr}\left(\Sigma^{-1}\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}\right) \\
&= c^{n_1+n_2-3}\operatorname{tr}\left(\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}\right) \\
&= \frac{\frac{2\sigma_2}{\sigma_1^2}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} c^{n_1+n_2-3}
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_2} &= \operatorname{tr}(\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_2}) \\
&= \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} \Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}\right) - \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w}, \Sigma)\frac{\partial \Sigma}{\partial \sigma_2}\right) \\
&= O_p(c^{n_1+n_2-3})
\end{aligned}
$$

Then we put $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \mu_1}$, $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \mu_2}$, $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_1}$, and $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_2}$ together and based on the definition of root sample-size consistent estimator,

$$
|\hat{\mu}_1 - \mu_1| = O_p(n_1^{-1/2}),
$$

$$
|\hat{\mu}_2 - \mu_2| = O_p(n_2^{-1/2}),
$$

$$
|\hat{\sigma}_1 - \sigma_1| = O_p(n_1^{-1/2}),
$$

$$|\hat{\sigma}_2 - \sigma_2| = O_p(n_2^{-1/2}),$$

we have

$$\begin{aligned}\Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma) \approx & \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}(\hat{\mu_1} - \mu_1) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_2}(\hat{\mu_2} - \mu_2) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_1}(\hat{\sigma_1} - \sigma_1) \\ & + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_2}(\hat{\sigma_2} - \sigma_2) \\ & = O_P(n_1^{\frac{1}{2}} c^{n_1+n_2-3})\end{aligned}$$

Finally,

$$\begin{aligned}&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \hat{\mu_1}.\hat{\mu_2}, \hat{\sigma_1}, \hat{\sigma_2}\right) - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \mu_1.\mu_2, \sigma_1, \sigma_2\right) \\ &= O_P(n_1^{\frac{1}{2}} n_2 c^{n_1+n_2-3} + n_1 n_2^{\frac{1}{2}} c^{n_1+n_2-3})\end{aligned}$$

□

3 Simulation Results for Theorem 4.3

We now show the simulation results of our Normal Distribution example under different configurations of $(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$. We verify Theorem 4.3 in the numerical simulation. Directly paradox probabilities from these theorems are compared with Monte Carlo paradox probabilities in a variety of settings. Each of these settings will involve comparisons of the observed statistics and latent aptitudes of the highest achievers, the pair $(Y_{i,(n_i)}, X_{i,(N_i)})$ across two populations.

The simulations will proceed as follows: Distributions for Y_1 and Y_2 are specified along with sample sizes n_i, N_i, $i = 1, 2$. We will specify a Monte Carlo sample size of 2000. For iteration m of the Monte Carlo sample procedure $Y_{1,j}$, $j = 1, \ldots, n_1$ and $Y_{2,k}$, $k = 1, \ldots, n_2$ are generated. Then $X_{i,(N_i)}$, $i = 1, 2$ is calculated using (2) where $X \sim \text{Pareto}(\alpha = 1.16)$. Then $1(X_{(N_1)} > X_{(N_2)}, Y_{(n_1)} < Y_{(n_2)})$ is recorded.

The theoretical results are computed from the Theorem 4.3 using mvtnorm package [Genz and Bretz, 2009] in R

The results are tabulated in Table 1. These results show strong agreement between our theory and numerical simulation. They also reveal that paradox probabilities are non-trivial across these configurations.

References

Alan Genz and Frank Bretz. *Computation of Multivariate Normal and t Probabilities*. Lecture Notes in Statistics. Springer-Verlag, Heidelberg, 2009. ISBN 978-3-642-01688-2.

n_1	n_2	N_1	N_2	μ_1	μ_2	σ_1	σ_2	Simulation Prob	Theoretical Prob
10	40	400	600	-1	1	1.0	3	0.4000	0.4000099
10	20	400	600	0	0	1.0	3	0.3970	0.3970305
10	40	400	300	-1	1	1.0	3	0.5715	0.5714174
10	20	400	600	0	1	1.5	2	0.3645	0.3647783
10	40	400	600	0	1	1.0	3	0.3995	0.3999245
20	40	400	600	0	1	1.0	3	0.3995	0.3999803
10	40	400	600	-1	0	1.5	3	0.4000	0.3995149
20	40	400	600	0	0	1.5	2	0.2970	0.2962431
20	40	400	600	0	0	1.0	3	0.3750	0.3740938
10	20	400	600	-1	1	1.0	3	0.3990	0.4000059

Table 1: Simulation Results for Normal Example.

Kaare Brandt Petersen, Michael Syskind Pedersen, et al. The matrix cookbook. *Technical University of Denmark*, 7(15):510, 2008.

Fuzhen Zhang. *The Schur complement and its applications*, volume 4. Springer Science & Business Media, 2006.